14 Volt Power Converter System – General Data

1. General

There are two 14 volt power outlets installed in the cockpit, on the pilot and co-pilot's side of center pedestal, to provide power to peripherals; cell phone, laptop, etc.

2. Description

Two 14 volt convenience outlets are provided in the flight compartment. They are located on either side of the instrument panel center pedestal, just below the base of the instrument panel. These two outlets have maximum combined load of 7.5 amps. The 28 Volt DC to 14 Volt DC power converter is located in the forward avionics compartment, and is mounted to the underside of the middle avionics shelf.

28 Volt DC power is provided from the right bus through a 5 amp PWR CONV 28V INPUT (M6) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data). 14 Volt DC power is provided to the outlets through a 7.5 amp PWR CONV 14V OUTPUT (N6) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

On aircraft with Forward Cabin 14V convenience outlets (Mod S.O.O. 6282) incorporated (refer to 25-22-00, 14 Volt Power System (Mod S.O.O. 6272, 6281 and 6282) – General data), 28 Volt DC power is provided through the right bus through a 20 amp PWR CONV 28V INPUT (M6) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data). 14 Volt DC power is provided to the outlets through a 7.5 amp PWR CONV 14V OUTPUT (N6) circuit breaker on the avionics circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

3. Adjustment/Test

A. Function Test of 14 Volt Power Converter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Using a voltmeter, measure the voltage between the center pin and the outer ring of the outlet. Ensure the 'com' lead is connected to the outer ring. Confirm the voltage is +13.8V (± 0.5V).

3) Plug an accessory into each outlet, as required, and ensure the accessory is being powered by the outlet.

4) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Adjustment Test of 14 Volt Power Converter (Post Mod 6/2406)

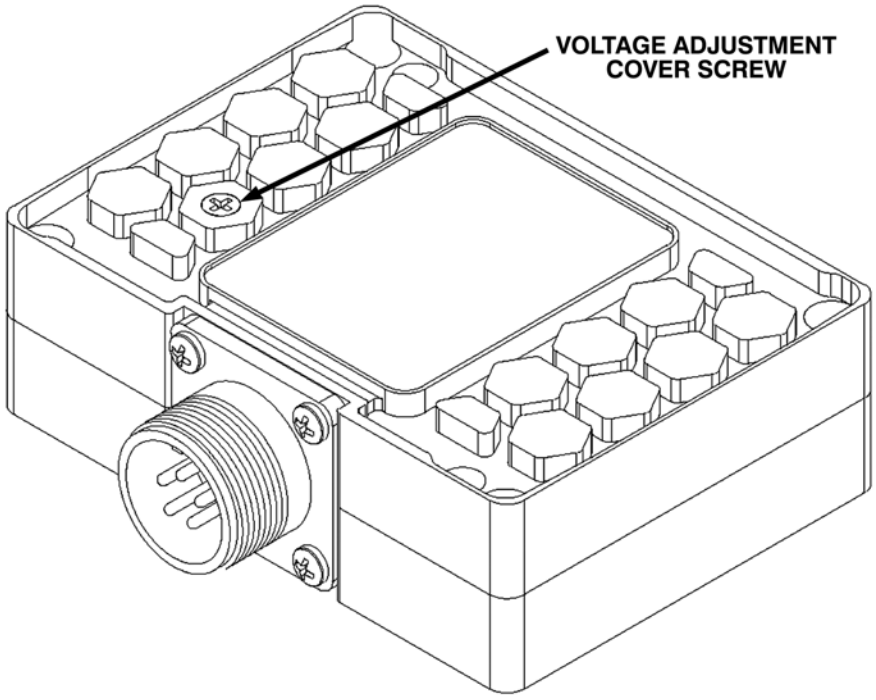
1) Disconnect any electrical load from outlet plug.

2) Remove voltage adjustment cover screw. (Refer to Figure 1).

3) Using a small flat screwdriver, adjust voltage trimmer clockwise to increase voltage or counterclockwise to decrease voltage.

4) Function test in accordance with Para 3.A.

5) Replace voltage adjustment cover screw after voltage adjustment is complete.



14V Power Converter (Post Mod 6/2406)
Figure 1

4. Removal/Installation

A. Removal of 14 Volt Power Converter

1) Ensure the aircraft electrical system is OFF.

2) Open the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel and fit with safety clips.

3) Gain access to the forward avionics shelves through the forward baggage compartment.

4) Remove the middle and bottom covers attached to bulkhead at station 44.00 as required.

5) Disconnect electrical connector.

6) While supporting the 14 volt power converter, remove the four mounting screws.

7) Remove unit from shelf.

Note Stow connectors as required.

B. Installation of 14 Volt Power Converter

1) Ensure the aircraft electrical system is OFF.

2) Ensure the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel is opened and fitted with safety clips.

3) Set unit in position and install four mounting screws.

4) Reconnect electrical connector.

5) Install the middle and bottom covers attached to bulkhead at station 44.00. Ensure the covers are mounted securely.

6) Remove the safety clips and close the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel.

7) Using a voltmeter, measure the voltage between the center pin and the outer ring of the outlet. Ensure the 'com' lead is connected to the outer ring. Confirm the voltage is +13.8V (± 0.5V).

8) Do a function test, refer to Para 3.A.

C. 14 Volt Power Outlet

1) Removal of 14 Volt Power Outlet

a) Ensure the aircraft electrical system is OFF.

b) Open the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel and fit safety clips.

c) Remove side panel(s) of center pedestal as required to access the rear of the 14 volt outlet.

d) Gently remove electrical connectors from the rear terminals of the outlet. Note which connector came off the center terminal and which came off the outer terminal.

e) Unthread the two metallic locknuts from the back of the outlet.

f) Remove the outlet by pulling it out the front of the mounting hole.

Note Stow connectors as required.

2) Installation of 14 Volt Power Outlet

a) Ensure the aircraft electrical system is OFF.

b) Ensure the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel is opened and fitted with safety clips.

c) Install the outlet in the mounting hole from the front of the panel.

d) Thread the two metallic locknuts onto the back of the outlet and tighten. Ensure outlet is secured in the panel.

e) Install electrical disconnects onto the appropriate leads on the rear of outlet.

f) Install side panels as required.

g) Remove the safety clips and close the PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers on the avionics circuit breaker panel.

h) Using a voltmeter, measure the voltage between the center pin and the outer ring of the outlet. Ensure the 'com' lead is connected to the outer ring. Confirm the voltage is +13.8V (± 0.5V).

i) Do a function test, refer to Para 3.A.

5. Troubleshooting

A. 14 Volt Power Outlet

1) Ensure PWR CONV 28V INPUT (M6) and PWR CONV 14V OUTPUT (N6) circuit breakers are closed.

2) Visually inspect the outlet connectors for damaged wiring or contacts. Ensure electrical connectors on the outlets are fully seated and secure.

3) Visually inspect the converter connector for damaged wiring or contacts. Ensure electrical connector on the converter is fully seated and secure.

4) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

5) Measure 14V output voltage at PWR CONV.

6) If repair or service is required, send to appropriately rated avionics repair facility.